

MLOps for ML engineers

Vladislav Goncharenko

Program

- Reproducibility in ML
 - What do we want to control?
 - CRISP DM
 - TDSP (Microsoft)
- Definition of MLOps and DevOps
- MLOps subtasks
- ML infrastructure in Big Tech
 - Amazon, Google, Microsoft, Nvidia
 - DataBricks, Seldon
- Problems of open source stack
- Packaging in Python
- Docker containers
 - Difference between Virtual Machines (VMs) and containers
- Environment management tools. Examples and their differences.
- Dependency management tools. Examples and their differences.
- Code quality tools
 - pre-commit
 - formatters
 - linters
 - other tools
- CLI instruments in Python
 - Fire
- Code writing techniques
 - SOLID
- Version Control Systems
 - Taxonomy and examples
- Git servers
- Workflow
 - Gitflow and friends
 - Git rebase
 -
- Git servers
- CI/CD stacks. Examples, differences
- Data sources
 - Data types taxonomy
- Data storages
 - Small data

- Big data
- Experiments parametrization
 - Hydra
- Logging
 - Examples, differences
- Neural network training frameworks
 - Examples, differences
- Contours of computation
 - Offline, online, etc
 - Their properties
- Computation models
- Metrics for computation speed
- Job runners
- What every programmer should know about memory
- GPU architecture
 - GPU vs CPU
 - von Neumann architecture
 - Flynn's taxonomy
 - FP32, FP16, INT8 computation performance
 - Arithmetic intensity (aka operational intensity)
- Automated mixed precision
 - Loss scaling
- Data annotation
 - Task management and roles
 - Assessor lifecycle
 - Honeypots
 - Overlap and aggregation
 - Tools
 - FOSS, proprietary